

AXIOM Project Brief

Open Rust Layer-1 validator and protocol-node hardening effort for trustworthy peer-to-peer infrastructure.

Project Summary

AXIOM is an open-source Rust implementation of a Layer-1 validator/node stack focused on secure, reproducible, and resilient peer-to-peer protocol infrastructure. The project emphasizes deterministic state transitions, authenticated peer messaging, persistence, replay safety, checkpointing, and operational resilience.

Current Status

- 59 Rust tests passing
- cargo clippy --all-targets --all-features -- -D warnings passing
- 24-hour soak test completed with 0 recorded failures
- Commit certificates with WAL replay integrated
- Persistent restartable state implemented
- Equivocation evidence and slashing coverage implemented
- Authenticated peer-session and signed message-path tests implemented
- Deterministic state-transition coverage implemented
- Storage/checkpoint hardening components implemented

Key Milestones

- Deterministic state-transition hardening in Rust
- Authenticated peer-session networking and stricter protocol enforcement
- Production-grade persistence improvements
- Snapshotting, checkpointing, and crash-recovery hardening
- Expanded fuzzing, abuse testing, load testing, and property-based testing
- Preparation for independent security review and public testnet

Repository

GitHub: <https://github.com/DS-DS-ship-it/axiom>

Audit target tag: v0.1.0-preaudit-hardening

Why NLnet Funding Helps

NLnet funding would help AXIOM continue as open-source internet/public-infrastructure work rather than a speculative token effort. The funding would support protocol hardening, authenticated peer networking, stronger persistence and checkpointing, reproducible testing, documentation, and preparation for independent security review and public testnet work.

The goal is to contribute a hardened, auditable, and reusable open Rust protocol/node implementation that others can study, test, and build on.

Prepared for grant, review, and infrastructure-support discussions.